STA2020F Tutorial on Logistic Regression

Question 1

1) South Africa is home to 12 species of tortoise, most of which are endangered through habitat loss. CapeNature therefore needs to monitor these species. Different species of tortoises can be told apart based on their appearance. But it is difficult to tell apart the sexes within each species. Researchers collected data on angulate tortoises (*Chersina angulata*) and in this tutorial, we want to see whether we can use morphological measurements to tell whether an individual is female or male. There were 242 tortoises in total. Of these, 99 were females (coded 1) and 143 males (coded 0).

   a) We first used the logistic regression model below to examine whether length (in mm) was a good predictor for sex in this species of tortoise.

      m1:

$$Y_i \sim Binom(1, p_i)$$

$$logit(p_i) = \beta_0 + \beta_1 \times length$$

Coefficients:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	22.72498	2.94934	7.705	1.31e-14
Length	-0.12296	0.01578	-7.793	6.54e-15

(Dispersion parameter for binomial family taken to be 1)
Null deviance: 327.44 on 241 degrees of freedom
Residual deviance: 193.53 on 240 degrees of freedom
AIC: 197.53

   i) What is the effect of length on the log odds of an individual being female (round to two decimal points)?

   ii) By what factor do the odds of being a female change for each unit increase in length?

   iii) Are longer tortoises relatively more likely to be males or females, compared to short tortoises?

   iv) what is the null hypothesis tested in the line starting with 'Length'?

   v) What do you conclude from the test?

   vi) Suppose that you have a tortoise of length 200mm. What is the probability that it is a female?

   b) From the model in question a), we predicted the sex of each tortoise based on its length. Using a threshold of p=0.5, we obtained the following confusion matrix:

	obs	
pred	0	1
0	126	22
1	17	77

i) How many tortoises were correctly predicted to be male?
ii) For how many tortoises did the model predict sex incorrectly?

iii) The results of the confusion matrix depend on the classification threshold chosen. True / False?

iv) What is the sensitivity of this classification tool?

c) Now, we fit two more models, m2 and m3 and obtain the following output:

m2:

$$Y_i \sim Binom(1, p_i)$$
$$logit(p_i) = \beta_0 + \beta_1 \times length + \beta_2 \times weight$$

Coefficients:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	36.193011	4.588543	7.888	3.08e-15
Weight	0.015496	0.002586	5.992	2.07e-09
Length	-0.280199	0.036273	-7.725	1.12e-14

m3:

$$Y_i \sim Binom(1, p_i)$$
$$logit(p_i) = \beta_0 + \beta_1 \times length + \beta_2 \times weight + \beta_3 \times length \times weight$$

Coefficients:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-1.418e+01	1.384e+01	-1.025	0.305542
Weight	7.263e-02	1.733e-02	4.191	2.78e-05
Length	-1.067e-02	7.791e-02	-0.137	0.891063
Weight:Length	-3.031e-04	8.956e-05	-3.385	0.000712

i) Based on model m2, what are the log odds of an individual with weight = 900g and length = 180mm to be a female?

ii) Based on model m2, what is the probability of an individual with weight = 900g and length = 180mm to be a female?

iii) Based on model m3, what is the probability of an individual with weight = 900g and length = 180mm to be a female?

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Question 2

1) Outline the following stepwise procedures:
   a) Forward selection
   b) Backward elimination
2) Discuss two problems with stepwise procedures
3) Compare and contrast R^2, Adjusted R^2, and AIC as criteria for model selection
4) Data is recorded to predict the price of a car based on numerous factors. In total, there are 24 variables. In order to reduce the number of variables in the model, stepwise procedures are considered. Two steps of an algorithm are provided.
   a) Which stepwise procedure was performed?
   b) What is happening between the first step and the second step?

```
Step:  AIC=2939.69
price ~ CarName + curbweight + enginetype + carwidth + carbody +
    peakrpm + aspiration + fuelsystem + carlength + stroke +
    doornumber + cylindernumber

                   Df Sum of Sq      RSS    AIC
+ symboling         1   2822461 65087125 2933.0
+ boreratio         1   2769493 65140093 2933.2
+ wheelbase         1   2753098 65156488 2933.2
+ citympg           1   1885568 66024018 2935.9
+ highwaympg        1   1566435 66343151 2936.9
+ compressionratio  1   1263478 66646108 2937.8
+ enginesize        1    847875 67061711 2939.1
<none>                          67909586 2939.7
+ horsepower        1    412171 67497414 2940.4
+ carheight         1    156333 67753253 2941.2
+ drivewheel        2     29175 67880411 2943.6

Step:  AIC=2932.99
price ~ CarName + curbweight + enginetype + carwidth + carbody +
    peakrpm + aspiration + fuelsystem + carlength + stroke +
    doornumber + cylindernumber + symboling

                   Df Sum of Sq      RSS    AIC
+ boreratio         1   5226332 59860793 2917.8
+ wheelbase         1   3876107 61211017 2922.4
+ compressionratio  1   2708235 62378889 2926.3
+ citympg           1   1931643 63155482 2928.8
+ highwaympg        1   1119190 63967935 2931.4
<none>                          65087125 2933.0
+ horsepower        1    414460 64672665 2933.7
+ carheight         1     87838 64999287 2934.7
+ enginesize        1      9867 65077258 2935.0
+ drivewheel        2     48577 65038548 2936.8
```

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Some extra practice

1) Students in UCT's STA2020S class were asked if they have ever driven after drinking. They were also asked how many days per month they drink at least two beers. Let *p* be the probability a student *i* said "yes" they had driven after drinking modelled against x_i, the number of days per month of drinking atleast 2 beers. Regression results are shown below:

```
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.5514     0.2661    -5.83   0.000
## DaysBeer       0.19031    0.02946    6.46   0.000
```

a) Using the regression output, calculate a, b, c, and d and complete the table below providing the estimated probabilities of ever having driven under the influence (*p*) for the given days per month of drinking beer.

DaysBeer	4	12	20	28
Predicted p	a	b	c	d

c) When DaysBeer is 10, what is the predicted odds of ever driving after drinking?

d) What is the predicted change in the log-odds of p associated with an increase of 5 days in number of days per month someone drinks more than 2 beers?

e) Now the sex (coded 1 for male and 0 for female) is included as an x variable along with DaysBeer.

```
## Coefficients:
##                Estimate   Std. Error   z value  Pr(>|z|)
## (Intercept)   -1.7736      0.2945       -6.02    0.000
##  DaysBeer      0.18693     0.03004       6.22    0.000
##  SexMale       0.6172      0.2954        2.09    0.037
```

What is the odds ratio of ever having driven after drinking when comparing males to females?

f) What is the predicted probability of ever having driven after drinking for a female who drinks at least 2 beers a day for 15 days of the month?

g) What is the predicted probability of ever having driven after drinking for a male who drinks at least 2 beers a day for 10 days of the month?

h) Interpret the meaning of intercept given in the model shown in part e? What is the meaning of $e^{\widehat{\beta}_0}$ for this model?

b) An observed binary outcome takes on categories "positive" and "negative" and a logistic regression model is run on an explanatory variable in order to assess how well the variable predicts this outcome.

		Observed	
		Positive	Negative
Predicted	Positive	20	33
	Negative	10	37

The model estimates a predicted probability p and based on the criteria "predicted outcome is positive if p is greater than 0.3", predicted outcomes are assigned. These predictions are compared to the actual observed outcomes as shown in the table above. For this model and this criteria for p, calculate the implied:

(a) Sensitivity

(b) Specificity

(c) positive predictive value

(d) negative predictive value